

Module 4: Bootstrap Framework

1. Introduction to Bootstrap

Bootstrap is a powerful, open-source frontend toolkit used for fast-paced web development. It provides a collection of pre-built CSS classes and JavaScript plugins that allow developers to create modern, professional interfaces without writing complex styles from scratch.

2. Advantages of Bootstrap

- **Speed of Development:** Ready-to-use components significantly reduce coding time.
- **Responsiveness:** Mobile-first approach ensures applications work across desktops, tablets, and handheld devices.
- **Consistency:** Provides a uniform look and feel across different browsers and platforms.
- **Extensive Documentation:** Offers comprehensive guides and a large community for support.

3. Bootstrap Grid System

The Grid system is a 12-column responsive layout system used to arrange content. It utilizes rows and columns to create complex layouts that adapt to different screen sizes.

- Example Code: `<div class="row">`

```
<div class="col-md-8">Main Content (8 columns)</div>
```

```
<div class="col-md-4">Sidebar (4 columns)</div>
```

```
</div>
```

- Screenshot Description: A browser window showing a wide horizontal section divided into two parts; the left part occupies roughly two-thirds of the width, while the right part occupies the remaining one-third.

4. Containers

Containers are the most basic layout element in Bootstrap, required when using the grid system.

- `.container`: Provides a responsive fixed-width container.
- `.container-fluid`: Provides a full-width container spanning the entire viewport width.

5. Buttons

Bootstrap provides several pre-styled button classes for different actions.

Class	Description
<code>.btn-primary</code>	Main action button (usually blue)
<code>.btn-secondary</code>	Secondary action (usually grey)
<code>.btn-success</code>	Positive action (usually green)
<code>.btn-danger</code>	Critical/Delete action (usually red)

- Screenshot Description: A row of rounded rectangular buttons in various colors (blue, grey, green, red) with white text inside.

6. Forms

Forms are essential for data entry portals. Bootstrap styles common form elements like `<input>`, `<label>`, and `<select>`.

- **Example Code:**`<div class="mb-3">`

`<label class="form-label">Email address</label>`

`<input type="email" class="form-control"
placeholder="name@example.com">`

`</div>`

- **Screenshot Description:** A clean input field with a light grey border that glows blue when clicked, accompanied by a clear text label above it.

7. Tables

Bootstrap tables are pre-styled with padding and light dividers. You can add classes like `.table-striped` for zebra-stripping or `.table-hover` to highlight rows on mouseover.

- **Screenshot Description:** A clean table with horizontal lines separating rows, where every second row has a slightly darker background for better readability.

8. Cards

Cards provide a flexible content container with options for headers, footers, and a wide variety of content.

- **Example (BHEL Asset Card):**`<div class="card shadow-sm">`

`<div class="card-body">`

`<h5 class="card-title text-primary">Add New Turbine Unit</h5>`

`<button class="btn btn-primary">Register Asset</button>`

</div>

</div>

- **Screenshot Description:** A white rectangular box with a subtle shadow around its edges, containing a blue title and a blue action button.

9. Navigation Bar

Navbars are responsive meta-components that serve as navigation headers for your application. They can collapse on smaller screens and expand on larger ones.

- **Screenshot Description:** A dark horizontal bar at the top of a page containing a brand logo on the left and a series of text links (e.g., Home, Features, Pricing) on the right.

10. Responsive Design using Bootstrap

Bootstrap uses a "mobile-first" approach. This means styles are applied to small devices first, and then modified for larger screens using media queries. The 12-column grid system is the primary tool for achieving this responsiveness.

- **Screenshot Description:** Two side-by-side images. The first shows a desktop view with three columns of content. The second shows a mobile view where those same three sections are stacked vertically to fit the narrow screen.

Congratulations! You have completed Module 4.